

[HKU AI+BIM Team] — Technical Assessment Invitation

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Dear candidate,

Thank you for your interest in the positions with our HKU AI+BIM team! This is a brief capability test designed to understand your AI-native learning velocity and engineering taste quickly. Please complete it within 7 days.

Task

Build a web-based micro-prototype or an intelligent agent within 7 days to perform basic compliance and sanity checks on architectural models/designs (implementing just 1–2 rules is sufficient).

Submission

- **Submission Deliverables:** Two links: 1) A GitHub repository (containing code + prompts); 2) A link to a demonstration/walkthrough video under 3 minutes.
- **Submission Method:** Email your submission to [\[junnaifj@hku.hk\]](mailto:junnaifj@hku.hk) with the subject line formatted as "【HKU AI Agent Technical Test】 Name_University".
- **Submission Deadline:** All deliverables must be submitted by **11:59 pm on 20 July 2026**.

Hints & Suggestions

1. **Audit Scope:** Please leverage AI to research BIM/CAD model checking, IFC structures, and engineering regulations (clash detection, fire safety, quantity takeoff, etc.). Implement only 1–2 rules—prioritise conciseness. For example: geometric clashes (walls/beams/pipes), clear exit door width checks, room-to-exit travel distances, or model property enrichment (e.g., names/FireRating).
2. **Data Formats:** Authentic IFC models, CAD drawings, simplified JSON representations, or AutoCAD/Revit exports are all acceptable. Please research and source your own samples, models, or use AI to generate test datasets.
3. **Implementation Form:** Web tools, intelligent Agents, or local scripts are all welcome. You have absolute freedom regarding your tech stack (e.g., Python+ifcopenshell, JS, LangChain/Claude Agent, etc.).
4. **Evaluation Focus:** Do not waste time chasing perfection; the core metric is "functional + thoughtful + tasteful". We look for clean code structure, practical and user-friendly human-AI interaction, sound engineering judgement behind design choices, and genuinely helpful data visualisation.
5. **Other Notes:** We place a far higher premium on your learning velocity, engineering judgement, and potential over pre-existing architecture domain knowledge.

中文说明:

感谢您对我们 HKU AI+BIM 团队职位的兴趣！这是一个简短的能力测试，旨在快速了解您利用 AI 的学习速度与工程品味。请在 7 天内完成。

测试任务

在7天内构建一个 Web 微原型 或 智能 Agent，用于对建筑模型/设计进行基础合规与合理性检查（实现 1-2条规则即可）。

提交

- 提交内容：两个链接，1) GitHub 仓库（代码 + 提示词）；2) 3分钟以内的演示/介绍视频的链接。
- 提交方式：按“【HKU AI Agent 笔试测试】姓名_学校”的主题形式通过邮件发送到 [\[junnaifj@hku.hk\]](mailto:junnaifj@hku.hk)。

提示与建议

1. 检查内容：请自行利用 AI 调研 BIM/CAD 模型检查、IFC 结构及规范（碰撞、消防、算量等）。只做1-2条规则，精炼优先。例如：几何碰撞（墙/梁/管道）、疏散门净宽度检查、房间到出口距离、模型属性完善（如名称/FireRating）等。
2. 数据格式：IFC 模型，CAD模型，JSON 简化模型，AutoCAD/Revit 导出等均可。请自行调研搜集 sample、模型或用 AI 生成测试数据。
3. 实现形式：Web 工具、智能 Agent、本地脚本均可。技术栈完全自由（Python+ifcopenshell、JS、LangChain/Claude Agent 等）。
4. 考察重点：无需追求完美，核心是“可运行 + 有思考 + 有品味”。代码结构清晰、人机交互友好实际、设计决策有判断力、结果可视化有帮助等。
5. 其他：我们更看重您的学习速度、工程判断力和潜力，而非已有建筑领域知识。

Please reply to this email to confirm your participation and submission timeline, and please contact us via email, junnaifj@hku.hk, if you have any questions regarding the process. We look forward to seeing your innovative work!

Best regards,

Jia

Research Assistant

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